PROJECT: ANALYZING INDUSTRY CARBON EMISSIONS

datalab

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When factoring heat generation required for the manufacturing and transportation of products, *Greenhouse gas emissions attributable to products, from food to sneakers to appliances, make up more than 75% of global emissions.* (`Source: The Carbon Catalogue https://www.nature.com/articles/s41597-022-01178-9`)

Our data, which is publicly available on nature.com, contains product carbon footprints (PCFs) for various companies. PCFs are the greenhouse gas emissions attributable to a given product, measured in CO_2 (carbon dioxide equivalent).

This data is stored in a PostgreSQL database containing one table, `product_emissions`, which looks at PCFs by product as well as the stage of production that these emissions occurred. Here's a snapshot of what `product_emissions` contains in each column:

`product_emissions`

field	data type
`id`	`VARCHAR`
`year`	`INT`
`product_name`	`VARCHAR`
`company`	`VARCHAR`
`country`	`VARCHAR`
`industry_group`	`VARCHAR`
`weight_kg`	`NUMERIC`
`carbon_footprint_pcf`	`NUMERIC`
`upstream_percent_total_pcf`	`VARCHAR`
`operations_percent_total_pcf`	`VARCHAR`

field	data type
`downstream_percent_total_pcf`	`VARCHAR`

You'll use this data to examine the carbon footprint of each industry in the dataset!

Projects Data DataFrame as `carbon_emissions_by_industry`

```sql
-- Update your query here
SELECT industry_group,
    COUNT(DISTINCT company) as num_companies,
    ROUND(SUM(carbon_footprint_pcf), 1) as total_industry_footprint
FROM product_emissions
    WHERE year IN (SELECT MIN (2017) FROM product_emissions)
GROUP BY industry_group
ORDER BY total_industry_footprint DESC
```

index	industry_group	num_companies	total_industry_footprint
0	Materials	3	107129
1	Capital Goods	2	94942.7
2	Technology Hardware & Equipment	4	21865.1
3	Food, Beverage & Tobacco	1	3161.5
4	Commercial & Professional Services	1	740.6
5	Software & Services	1	690

Rows: 6 Expand